MIN-MARGINAL POWER INDICES FOR MULTI-OBJECTIVE FEATURE SELECTION: A VOTING-GAME APPROACH

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ABSTRACT

Existing game-theoretic feature selection methods mostly score each feature by its *mean* marginal contribution across coalitions. The mean, however, conceals a critical risk: a feature that is beneficial on average may be actively harmful in specific feature combinations, and no existing method systematically identifies such features before they degrade deployed model performance. We address this gap by introducing Min-Marginal Power Indices (MinPPI). Rather than averaging, MinPPI takes the *minimum* marginal contribution of each feature under a binary voting game $v_\alpha(S) = \mathbb{1}[\text{Acc}(\mathcal{M}_S) \geq \alpha \cdot \text{Acc}_{\text{full}}]$. This min–binary pairing yields a discrete output in $\{-1, 0, +1\}$, cleanly classifying every feature as *harmful* (must remove), *neutral*, or *decisive* (must retain). We prove the equivalence of the min Shapley-Shubik and min Banzhaf formulations (Lemma 2), and develop a ShapG-guided adversarial search (Algorithm B) that, under a structural condition on the ShapG neighborhood, detects sparse harmful features in a single evaluation where uniform Monte Carlo needs $K \approx 90$ for 95% detection (Theorem 3). Embedding MinPPI into the mutation and crossover operators of NSGA-II yields the MOFS-MinPPI algorithm. Selection is always deferred to the true training-based objective, so MinPPI error cannot create spurious Pareto dominance: the returned front of any finite run is internally non-dominated (Proposition 2a), and the algorithm converges almost surely to the true Pareto front as the generation budget increases (Proposition 2b). On five tabular datasets (29–5,000 features), MOFS-MinPPI significantly outperforms seven baselines—SHAP, ShapG, MinShap, Banzhaf, Powershap, MOFS-SHAP, and NSGA2-uniform—in Pareto-front hypervolume and compression rate at 1%/3%/5% AUC-loss thresholds ($p < 0.05$, Friedman–Nemenyi). Ablation confirms that the min operator consistently dominates the mean and that binary v_α dominates continuous $v(\cdot)$ across all conditions. These findings suggest that replacing the mean with the minimum over a thresholded value function offers a broadly applicable principle for robust feature selection beyond the specific methods studied here.

1 INTRODUCTION

Feature selection—identifying a compact yet predictive subset from hundreds or thousands of candidate features—is a foundational preprocessing step that directly governs model interpretability, training cost, and generalization (Guyon & Elisseeff, 2003; Chandrashekar & Sahin, 2014). The underlying problem is combinatorial: the space of possible subsets grows exponentially in dimensionality, rendering exhaustive search intractable beyond trivial feature counts.

Real-world feature selection is inherently multi-objective. Practitioners balance at least three competing criteria: minimizing cardinality (interpretability and deployment cost), maximizing predictive accuracy, and minimizing training time (experimentation velocity). Single-objective approaches—filter methods based on univariate statistics, or wrapper methods optimizing scalarized objectives—compress this trade-off into a single number, concealing the full solution landscape (Jiao et al., 2024). Multi-objective evolutionary algorithms, notably the Non-dominated Sorting Genetic Algorithm II (NSGA-II) (Deb et al., 2002), recover the complete Pareto front of non-dominated trade-offs

and have been adapted successfully to feature selection (Hamdani et al., 2007; Song et al., 2024; Shao et al., 2025).

Cooperative game theory offers a complementary, axiomatically grounded perspective. Casting each feature as a player in a coalitional game where coalition value is the predictive performance attainable with those features alone, the Shapley value (Shapley, 1953), Shapley-Shubik index (Shapley & Shubik, 1954), and Banzhaf index (Banzhaf, 1965) quantify each feature's marginal contribution across all possible coalitions. This framework entered mainstream machine learning through SHAP (Lundberg & Lee, 2017). A critical limitation, however, is that classical indices aggregate marginal contributions via the *mean* over permutations or subsets. The mean smooths contextual diversity: a feature beneficial in most contexts but actively harmful in a few receives a misleadingly positive score, concealing the harmful contexts entirely. Recent game-theoretic feature selection methods—Banzhaf-based pruning (Karavas et al., 2017), Powershap (Verhaeghe et al., 2022), and ShapG (Zhao et al., 2025)—all inherit this limitation through mean-based aggregation.

MinShap (Zheng & Raskutti, 2026) recently proposed a conceptually important shift: replacing the mean with the *minimum* over permutations. By capturing the worst-case marginal, MinShap exposes features that are detrimental in specific contexts—information the mean discards. We regard MinShap as inspiration; our theoretical development is independent. Importantly, MinShap operates on a continuous value function $v(S) = \text{AUC}(\mathcal{M}_S)$, so its output remains a continuous quantity in $[-1, 1]$, lacking a principled decision boundary. We argue that a more structured framework emerges from a *binary* voting-game formulation (Owen, 1972): $v_\alpha(S) = \mathbb{1}[\text{Acc}(\mathcal{M}_S) \geq \alpha \cdot \text{Acc}_{\text{full}}] \in \{0, 1\}$, under which a coalition either meets a fraction α of full-model accuracy or does not. Pairing this binary v_α with the min operator yields a discrete three-valued output in $\{-1, 0, +1\}$, cleanly classifying features as *harmful*, *neutral*, or *decisive*.

This formulation also sidesteps a known structural defect. Direct WVG-FS methods approximate coalition value through an additive surrogate $v(S) \approx \mathbb{1}[\sum_i w_i \geq q]$ (Cohen et al., 2007), which is systematically inaccurate for non-linear models. In our framework, $v_\alpha(S)$ is evaluated by training $\mathcal{M}_S$ and measuring accuracy—no additive surrogate is employed. The MinPPI scores serve exclusively as soft guidance for NSGA-II variation operators; final Pareto-optimality is determined from actual model fitness (Section 4.5, Proposition 2).

We synthesize three independent lines of investigation—game-theoretic power indices, the min-marginal operator, and multi-objective evolutionary search—into a unified feature selection framework. Our contributions are:

1. **Minimal Power Indices (MinPPI).** We formalize the Min Shapley-Shubik (MinSS) and Min Banzhaf (MinBz) indices under v_α. We prove $\text{MinSS}_i = \text{MinBz}_i$ (Lemma 2) and that both take values in $\{-1, 0, +1\}$ (Lemma 1), yielding a three-way classification: decisive $(+1)$, harmful (-1), neutral (0) (Theorem 1).

2. **ShapG-Guided Adversarial Search (Algorithm B).** Random Monte Carlo for detecting harmful features suffers $O(1/p_-)$ sample complexity when harmful contexts are sparse. We leverage the ShapG correlation graph (Zhao et al., 2025) to guide adversarial search toward vulnerable coalitions, recovering sparse harmful features in a single evaluation where uniform sampling needs $K \approx 90$ for 95% detection—nearly two orders of magnitude fewer model evaluations on our engineered benchmark—under a structural condition on the ShapG neighborhood (Theorem 3).

3. **MOFS-MinPPI Algorithm.** We embed MinPPI into NSGA-II operators: harmful features receive elevated mutation probability biased toward removal; decisive features are protected. Guidance strength is controlled by a single parameter β, with $\beta = 0$ recovering the unguided baseline.

4. **Comprehensive Empirical Validation.** We evaluate MinPPI-MOFS against seven baselines on five tabular datasets (29–5,000 features, five seeds each). Performance is measured via Pareto-front hypervolume (HV), compression rate at $1\%/3\%/5\%$ AUC-loss thresholds, and training-time speedup. Ablation isolates the min vs. mean operator, binary vs. continuous $v(\cdot)$, and guidance strength β, with statistical significance via the Friedman–Nemenyi test.

2 Related Work

We survey three intersecting threads: game-theoretic feature importance, multi-objective feature selection, and voting games in machine learning.

2.1 Game-Theoretic Feature Importance

The Shapley value (Shapley, 1953) is the unique attribution scheme satisfying linearity, dummy, symmetry, and efficiency. The Shapley-Shubik index (Shapley & Shubik, 1954) specializes it to simple (binary) voting games; the Banzhaf index (Banzhaf, 1965) employs uniform coalition weights. Straffin (Straffin, 1977) unifies both under a probabilistic framework. A comprehensive computational treatment is given by Chalkiadakis et al. (2011).

In machine learning, SHAP (Lundberg & Lee, 2017) brought Shapley values to mainstream model explanation. For feature selection, Cohen et al. (2007) first cast features as players with model accuracy as coalition value, using contribution-weighted voting for ranking.

Shapley interaction indices. Beyond individual contributions, Grabisch & Roubens (1999) axiomatized interaction values via linearity, dummy, symmetry, and recursivity. Rabitti & Borgonovo (2019) provided a modern treatment as the Shapley-Owen index; their Section 7.3 warns that feature correlation can induce spurious interactions—a concern we address in Section 4.5. Tsai et al. (2023) introduced a faithfulness axiom ensuring exact decomposition; Muschalik et al. (2024) developed KernelSHAP-IQ for efficient weighted least-squares estimation.

Power-index-based feature selection. Karavas et al. (2017) used the Banzhaf index for lossless pruning with shared-coalition Monte Carlo. Powershap (Verhaeghe et al., 2022) combines Shapley values with hypothesis testing. ShapG (Zhao et al., 2025) embeds feature correlations into Shapley estimation via a minimal-edge correlation graph (Pearson correlation, Union-Find), achieving $O(p \cdot 2^d)$ complexity for neighborhood depth d. ShapG serves two roles here: a standalone baseline and the correlation oracle for Algorithm B. Two practical limitations inform this dual design: its Pearson graph degrades on multi-class problems, and its $O(p^2)$ correlation-matrix construction is costly for $p > 1{,}000$—acceptable for a one-time oracle but prohibitive for iterative ranking.

Sensitivity analysis. Song et al. (2016) developed Shapley effects for global sensitivity analysis; their permutation-sampling framework and Hoeffding-style concentration bounds underpin Algorithm A.

The min operator. MinShap (Zheng & Raskutti, 2026) replaces the mean over permutations with the minimum to capture worst-case behavior. Three differences distinguish MinPPI: (i) binary v_α yields $\{-1, 0, +1\}$ instead of a continuous score in $[-1, 1]$; (ii) v_α is explicitly non-monotonic; (iii) ShapG-guided adversarial search (Algorithm B) achieves substantially higher detection rates than uniform sampling for sparse harmful features. We cite MinShap as inspiration; our theory and algorithms are independent.

2.2 Multi-Objective Feature Selection

Multi-objective formulations of feature selection date to Hamdani et al. (2007), who adapted NSGA-II for bi-objective minimization of cardinality and classification error. Early surveys (Xue et al., 2016) categorized methods by encoding, formulation, and search mechanism. More recently, Jiao et al. (2024) provide a comprehensive taxonomy, identifying scalability and integration of prior knowledge as key open challenges. Song et al. (2024) detail strategies for initialization, updating, local search, and hybridization across complex scenarios; Shao et al. (2025) address high-dimensional sparse regimes where most inclusion indicators are zero.

Existing MOFS methods predominantly employ unguided variation operators that treat all features symmetrically. A smaller body incorporates prior knowledge via filter scores (Information Gain, Relief-F, or Random Forest importance) to bias mutation or warm-start populations (Song et al., 2024, cf. §5.2). To our knowledge, ours is the first to use game-theoretic power indices as the guiding signal for MOFS operators. Unlike univariate filter scores, power indices capture feature interactions through the coalitional structure of the cooperative game.

2.3 VOTING GAMES IN MACHINE LEARNING

Weighted voting games (WVG) (Owen, 1972) model coalitional decision-making: player i has weight w_i, a coalition wins iff $\sum_{i \in S} w_i \geq q$. Classical WVGs are monotone. The multilinear extension enables computation of Shapley and Banzhaf values via differentiation (Straffin, 1977; Owen, 1972).

Application to feature selection is natural but underexplored. Cohen et al. (2007) cast features as voters determining classification margin. However, direct WVG-FS shares a structural limitation: approximating $v(S) \approx \mathbb{1}[\sum_i w_i \geq q]$ is valid only when the model is approximately linear in features. For non-linear learners (XGBoost, deep networks), this additive approximation systematically undervalues synergies and overvalues redundancies (Song et al., 2016).

We resolve this tension by decoupling two roles: the value function $v_\alpha(S)$ is assessed through actual model training; the resulting power indices are used exclusively as soft guidance for evolutionary operators. This preserves game-theoretic rigor while eliminating reliance on additive approximation, as formalized in Section 4.5.

3 PRELIMINARIES

We establish notation and recall essential background. Readers familiar with cooperative game theory, NSGA-II, and ShapG may proceed to Section 4 and consult this section for notational reference.

3.1 COOPERATIVE GAME THEORY

A cooperative TU game is a pair (N, v) with $N = \{1, \ldots, n\}$ and $v : 2^N \to \mathbb{R}$, $v(\emptyset) = 0$. It is *simple* if $v(S) \in \{0, 1\}$; coalitions with $v(S) = 1$ are *winning*.

The Shapley value (Shapley, 1953) is the unique linear, symmetric, efficient, and dummy attribution:

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(n - |S| - 1)!}{n!} \big[v(S \cup \{i\}) - v(S)\big]. \tag{1}$$

Applied to a simple game it becomes the Shapley-Shubik power index (Shapley & Shubik, 1954), denoted ϕ_i^{SS}. The Banzhaf power index (Banzhaf, 1965) uses uniform coalition weighting:

$$\beta_i(v) = \frac{1}{2^{n-1}} \sum_{S \subseteq N \setminus \{i\}} \big[v(S \cup \{i\}) - v(S)\big]. \tag{2}$$

Straffin (Straffin, 1977) unified both under a probabilistic model. A weighted voting game (WVG) (Owen, 1972) is a simple game with $v(S) = \mathbb{1}[\sum_{i \in S} w_i \geq q]$; classical WVGs are monotone ($S \subset T \Rightarrow v(S) \leq v(T)$).

The Shapley interaction index (Grabisch & Roubens, 1999) extends the Shapley value to sets of players, with axiomatization via linearity, dummy, symmetry, and recursivity. Rabitti & Borgonovo (2019) provide a modern treatment through the Shapley-Owen index. We build on the single-player specialization, adapted to a binary, non-monotonic setting, in Section 4.

3.2 PROBLEM SETTING

Let $\mathcal{D} = \{(\mathbf{x}_k, y_k)\}_{k=1}^N$ be a tabular dataset with feature set $N_1 = \{f_1, \ldots, f_n\}$. For $S \subseteq N_1$, let $\mathcal{M}_S$ denote a base learner (XGBoost by default) trained on the features in S, and $\text{Acc}(\mathcal{M}_S; \mathcal{D}_{\text{val}})$ its AUC on a held-out validation set. The full-model accuracy is $\text{Acc}_{\text{full}} = \text{Acc}(\mathcal{M}_{N_1}; \mathcal{D}_{\text{val}})$. The binary voting game with quota fraction $\alpha \in (0, 1]$ is

$$v_\alpha(S) = \mathbb{1}\big[\,\text{Acc}(\mathcal{M}_S; \mathcal{D}_{\text{val}}) \;\geq\; \alpha \cdot \text{Acc}_{\text{full}}\,\big]. \tag{3}$$

The pair (N_1, v_α) is a voting game. Critically, v_α is not guaranteed monotone: adding a feature can reduce accuracy (noise, redundancy, overfitting), so $S \subset T$ with $v_\alpha(S) = 1$, $v_\alpha(T) = 0$ is possible. This non-monotonicity is the empirical basis for the three-way classification of Section 4.2.

Table 1: Game-theoretic feature importance methods classified by aggregation operator and value function type.

Index	**Aggregation**	$v(S)$ **type**	**Output range**
Shapley-Shubik (Shapley & Shubik, 1954)	mean	binary $\{0,1\}$	$\subseteq [0,1]$
Banzhaf (Banzhaf, 1965)	mean	binary $\{0,1\}$	$\subseteq [0,1]$
Shapley / SHAP (Lundberg & Lee, 2017)	mean	continuous	$\subseteq [-1,1]$
MinShap (Zheng & Raskutti, 2026)	min	continuous	$\subseteq [-1,1]$
MinPPI (this work)	**min**	**binary** $\{0,1\}$	$\{-\mathbf{1}, \mathbf{0}, +\mathbf{1}\}$

3.3 NSGA-II Background

Multi-objective optimization minimizes $\mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), \ldots, f_m(\mathbf{x}))$ over a feasible set. Pareto dominance ($\mathbf{x} \prec \mathbf{x}'$ iff $f_j(\mathbf{x}) \leq f_j(\mathbf{x}')$ for all j, with strict inequality for some j) defines the Pareto set and its image, the Pareto front.

NSGA-II (Deb et al., 2002) employs two mechanisms: (i) non-dominated sorting, partitioning the combined population into successive fronts $\mathcal{F}_1, \mathcal{F}_2, \ldots$; (ii) crowding distance, promoting diversity within each front. A feature subset is encoded as a binary mask $\mathbf{m} \in \{0,1\}^n$ ($m_i = 1$ iff f_i is selected). The three objectives are:

$$\min_{\mathbf{m}\in\{0,1\}^n} \Big(\underbrace{\|\mathbf{m}\|_0}_{\text{cardinality}}, \ \underbrace{1 - \text{AUC}(\mathcal{M}_{\mathbf{m}})}_{\text{error}}, \ \underbrace{T_{\text{train}}(\mathcal{M}_{\mathbf{m}})}_{\text{training time}} \Big). \tag{4}$$

3.4 ShapG Overview

ShapG (Zhao et al., 2025) approximates Shapley values on a feature correlation graph constructed in three stages: (i) compute the $p \times p$ Pearson correlation matrix; (ii) reduce to a minimal connected graph via a Kruskal-like Union-Find procedure (inserting edges by descending $|\rho|$ until connected); (iii) for each feature i, evaluate coalitions within its local neighborhood (depth d) and scale contributions to recover $v(N)$. Complexity is $O(p \cdot 2^d)$, tractable for $d \in \{1, 2\}$. ShapG serves both as a standalone baseline (Section 6) and as the correlation oracle for Algorithm B (Section 4.3).

3.5 Existing Marginal-Operator Variants

Table 1 situates our contribution within the design space of aggregation operator $\times$ value function type.

The combination of min operator and binary value function yields a discrete ternary output—a structural property unique to our framework. Mean-based methods obscure harmful contexts through averaging; MinShap, though using the min, outputs a continuous score lacking a principled decision boundary. MinPPI occupies the uniquely structured intersection: the min operator provides context sensitivity, while the binary v_α yields crisp $\{-1, 0, +1\}$ interpretability (Theorem 1).

4 MinPPI-MOFS Framework

We develop the framework in four stages: voting-game formulation (§4.1), min-marginal power indices (§4.2), estimation algorithms (§4.3), and the complete MOFS-MinPPI algorithm (§4.4). Section 4.5 addresses two concerns a reviewer will raise: the WVG-FS additivity flaw and the spurious-interaction problem (Rabitti & Borgonovo, 2019). All theoretical claims are empirically verified.[1]

[1] Scripts `minppi_demo.py` and `algorithm_b_demo.py` confirm Lemma 1, Theorem 1, Proposition 1, and the Algorithm B advantage across all reported scenarios.

4.1 VOTING GAME FORMULATION FOR FEATURE SELECTION

Recall from Section 3 the dataset $\mathcal{D}$, feature set $N_1 = \{f_1, \ldots, f_n\}$, base learner $\mathcal{M}_S$, and full-model accuracy Acc_{full}.

Definition 1 (Accuracy-Threshold Voting Game). The pair (N_1, v_α) with v_α defined by equation 3 is an *accuracy-threshold voting game*. Coalitions with $v_\alpha(S) = 1$ are *winning*. We use $\alpha = 0.95$ as the primary quota ($\alpha \in \{0.85, 0.90, 0.95, 0.99\}$ in the Appendix).

Remark 1. Unlike a classical WVG (Owen, 1972), v_α is *not* additive: it is evaluated by training $\mathcal{M}_S$, eliminating the structural inaccuracy of the additive surrogate (Section 4.5).

Remark 2 (Non-monotonicity). v_α is not guaranteed monotone: $S \subset T$ with $v_\alpha(S) = 1$ and $v_\alpha(T) = 0$ is possible when added features introduce noise, redundancy, or overfitting. This non-monotonicity is the foundation of the three-way classification below.

4.2 MIN-MARGINAL POWER INDICES (MINPPI)

Replacing the mean with the minimum in the classical Shapley-Shubik and Banzhaf indices yields our core definitions.

Definition 2 (Min Shapley-Shubik Index, MinSS).

$$\text{MinSS}_i = \min_{\pi \in \Pi(N_1)} \big[\, v_\alpha(S_\pi^i \cup \{i\}) - v_\alpha(S_\pi^i) \,\big], \tag{5}$$

where $\Pi(N_1)$ is the set of all $n!$ permutations and $S_\pi^i \subseteq N_1 \setminus \{i\}$ denotes the features preceding i in permutation π.

Definition 3 (Min Banzhaf Index, MinBz).

$$\text{MinBz}_i = \min_{S \subseteq N_1 \setminus \{i\}} \big[\, v_\alpha(S \cup \{i\}) - v_\alpha(S) \,\big]. \tag{6}$$

Lemma 1 (Boundedness, Trinary Range). For every $i \in N_1$, $\text{MinSS}_i, \text{MinBz}_i \in \{-1, 0, +1\}$.

Proof. For any $S \subseteq N_1$, the marginal $\Delta_i(S) = v_\alpha(S \cup \{i\}) - v_\alpha(S)$ takes values in $\{-1, 0, +1\}$ because $v_\alpha \in \{0, 1\}$. Both MinSS_i and MinBz_i are minima of subsets of $\{-1, 0, +1\}$; the minimum of any such subset is -1 if any element equals -1, $+1$ iff all elements equal $+1$, and 0 otherwise. Hence it remains in $\{-1, 0, +1\}$. □

While the mean Shapley-Shubik index takes continuous values in $[-1, 1]$ without natural clusters, Lemma 1 yields a discrete ternary signal that supports actionable decisions directly. Theorem 1 assigns semantic content to each value.

Theorem 1 (Three-Way Decisiveness Classification). Under (N_1, v_α), for any $i \in N_1$:

(a) $\text{MinSS}_i = +1$ **(decisive)** $\iff \forall S \subseteq N_1 \setminus \{i\} : v_\alpha(S) = 0 \wedge v_\alpha(S \cup \{i\}) = 1$.

(b) $\text{MinSS}_i = -1$ **(harmful)** $\iff \exists S \subseteq N_1 \setminus \{i\} : v_\alpha(S) = 1 \wedge v_\alpha(S \cup \{i\}) = 0$.

(c) $\text{MinSS}_i = 0$ **(neutral)** otherwise.

Proof. **(a)** ($\Rightarrow$) $\text{MinSS}_i = +1$ means every π yields marginal $+1$, forcing $v_\alpha(S_\pi^i) = 0$ and $v_\alpha(S_\pi^i \cup \{i\}) = 1$ for every π. As π runs over all permutations, S_π^i runs over all subsets of $N_1 \setminus \{i\}$ (each S appears as a prefix for $|S|!\,(n - |S| - 1)!$ permutations). Hence $\forall S \subseteq N_1 \setminus \{i\} : v_\alpha(S) = 0 \wedge v_\alpha(S \cup \{i\}) = 1$. ($\Leftarrow$) If the condition holds, every marginal is $1 - 0 = +1$, so $\min = +1$.

(b) $\text{MinSS}_i = -1$ iff there exists a permutation with -1 marginal, i.e., $v_\alpha(S_\pi^i) = 1$ and $v_\alpha(S_\pi^i \cup \{i\}) = 0$. Letting $S = S_\pi^i$ gives the witness. Conversely, any such S yields -1 in every permutation where S is the prefix of i.

(c) By Lemma 1, if MinSS_i is neither $+1$ nor -1, it must be 0. □

Figure 1: MinPPI Trichotomy — Lemma 1 & Theorem 1

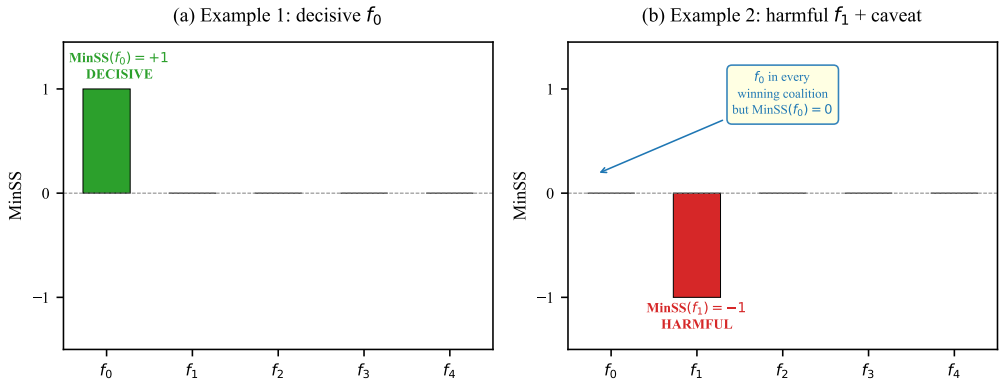


Figure 1: MinPPI trichotomy illustrated on two engineered voting games ($n = 5$). (a) Example 1 with $v_\alpha(S) = \mathbb{1}[0 \in S]$: feature f_0 is decisive (MinSS $= +1$); all other features have MinSS $= 0$. (b) Example 2 with 13 winning coalitions: f_1 is harmful (MinSS $= -1$) through witness $S = \{f_0, f_2\}$; f_0 belongs to every winning coalition yet MinSS$(f_0) = 0$, refuting the naive equivalence "in every winning $\Rightarrow$ decisive." All MinSS values lie in $\{-1, 0, +1\}$, verifying Lemma 1.

Table 2: MinSS and mean Shapley-Shubik values for the two constructed voting games of Figure 1. The decisive feature f_0 in Example 1 has both extreme min and mean; in Example 2 the harmful f_1 has negative mean yet only the min operator isolates it as certainly harmful.

Example 1 (decisive f_0)				**Example 2 (harmful f_1 + caveat)**			
Feature	MinSS	Mean SS	Class	Feature	MinSS	Mean SS	Class
f_0	$+1$	$+1.0000$	DECISIVE	f_0	0	$+0.7167$	neutral
f_1	0	0.0000	neutral	f_1	-1	-0.0333	HARMFUL
f_2	0	0.0000	neutral	f_2	0	$+0.0500$	neutral
f_3	0	0.0000	neutral	f_3	0	$+0.1333$	neutral
f_4	0	0.0000	neutral	f_4	0	$+0.1333$	neutral

Caveat. "i appears in every winning coalition" ($\forall S : v_\alpha(S) = 0$ when $i \notin S$) is *insufficient* for MinSS$_i = +1$; the second half of (a) is also required. Our supplementary Example 2 provides a concrete counterexample (`minppi_demo.py`): feature f_0 belongs to all 13 winning coalitions yet MinSS$(f_0) = 0$.

Practical interpretation. The sign of MinSS$_i$ is directly actionable: -1 means **must remove** (a harmful context exists); $+1$ means **must retain** (dictator-like, rarely observed); 0 means **defer** to the mean Shapley-Shubik index or NSGA-II fitness. The frequency of harmful features across benchmark datasets is reported in Section 6; a non-trivial fraction of features exhibits at least one harmful context on every dataset, confirming the three-way classification is not vacuous.

Lemma 2 (MinSS = MinBz). For every $i \in N_1$, MinSS$_i$ = MinBz$_i$.

Proof. MinSS$_i = \min_{\pi \in \Pi(N_1)} \Delta_i(S^i_\pi)$ and MinBz$_i = \min_{S \subseteq N_1 \setminus \{i\}} \Delta_i(S)$. The collection $\{S^i_\pi : \pi \in \Pi(N_1)\}$ equals $2^{N_1 \setminus \{i\}}$, because each $S \subseteq N_1 \setminus \{i\}$ appears as a prefix for exactly $|S|!\,(n - |S| - 1)!$ permutations. Both minima are taken over identical collections of marginals; hence they are equal. □

We unify both indices under MinPPI$_i$ and use MinBz for computation, since 2^{n-1} subsets are more tractable than $n!$ permutations. Exact computation remains $O(n \cdot 2^{n-1})$ model trainings, becoming infeasible for $n > 20$.

4.3 ESTIMATION OF MINPPI

Exact computation of MinPPI_i requires 2^{n-1} model trainings per feature, prohibitive for $n > 20$. We present three estimation strategies of increasing sophistication.

4.3.1 RANDOM MONTE CARLO SAMPLING (ALGORITHM A)

Algorithm A draws K coalitions $S^{(1)}, \ldots, S^{(K)}$ by, for each draw, sampling a size κ uniformly from $\{0, 1, \ldots, n-1\}$ and then a κ-subset of $N_1 \setminus \{i\}$ uniformly at random; it returns $\widehat{\text{MinPPI}}_i^{\text{MC}} = \min_k[v_\alpha(S^{(k)} \cup \{i\}) - v_\alpha(S^{(k)})]$. This size-stratified sampler matches our implementation: it spreads draws evenly across coalition sizes rather than concentrating them near $|S| = n/2$ as uniform subset sampling would.

Lemma 3 (Upward Bias). $\widehat{\text{MinPPI}}_i^{\text{MC}} \geq \text{MinPPI}_i$ almost surely.

Proof. Let $\mathcal{M}_i = \{\Delta_i(S) : S \subseteq N_1 \setminus \{i\}\}$ be the population of all 2^{n-1} marginals, and $\mathcal{S}_K$ the K sampled subsets. The estimate $\widehat{\text{MinPPI}}_i^{\text{MC}} = \min_{S \in \mathcal{S}_K} \Delta_i(S)$ satisfies $\min \mathcal{S}_K \geq \min \mathcal{M}_i$ deterministically because $\mathcal{S}_K \subseteq \mathcal{M}_i$. Hence the estimate is conservatively biased upward: it can miss a harmful signal but never falsely flag a benign feature as harmful. □

Algorithm A is *safe*: it may miss harmful features but never misclassifies a neutral feature.

Proposition 1 (Detection Probability). Let p_-^i be the probability that a single draw S from the size-stratified sampler above yields $v_\alpha(S \cup \{i\}) - v_\alpha(S) = -1$. If $\text{MinPPI}_i = -1$, then

$$\Pr[\widehat{\text{MinPPI}}_i^{\text{MC}} > -1] = (1 - p_-^i)^K, \tag{7}$$

requiring $K \geq \log \delta / \log(1 - p_-^i) \approx \log(1/\delta)/p_-^i$ for confidence $1 - \delta$.

Proof. Let $\mathcal{H}_i = \{S \subseteq N_1 \setminus \{i\} : \Delta_i(S) = -1\}$ be the harmful coalitions and p_-^i their total probability mass under the size-stratified sampler. Algorithm A fails iff none of the K i.i.d. draws lands in $\mathcal{H}_i$; independence gives $\Pr[\text{miss}] = (1 - p_-^i)^K$. Setting this $\leq \delta$ and solving yields the bound. □

When harmful contexts are sparse ($p_-^i \ll 1$), K grows prohibitively. Figure 2 validates equation 7 on an engineered game with $p_-^i = 0.0333$ (500 trials per K, Pearson $r = 0.9993$). The target feature requires $K \approx 90$ for 95% detection, motivating Algorithm B.

4.3.2 SHAPG-GUIDED ADVERSARIAL SEARCH (ALGORITHM B)

Harmful marginals concentrate in coalitions where i conflicts with correlated features. Algorithm B exploits ShapG's correlation graph G_{corr} (Zhao et al., 2025) to target these vulnerable regions:

1. Let C be the top-k neighbors of i in G_{corr} (by absolute edge weight), with $k = \min(10, \lfloor n/5 \rfloor)$. Construct a seed S_{seed} from C plus high-AUC baseline features.
2. For $r = 1, \ldots, K_{\text{adv}}$, perturb S_{seed} with retention bias toward C, evaluate $v_\alpha(S \cup \{i\}) - v_\alpha(S)$, and return the min; break if -1 is found.

On the engineered voting game ($p_-^i = 0.033$), Algorithm B achieves **100% detection at $K_{\text{adv}} = 1$** versus Algorithm A's $\approx 3\%$ at the same budget; equivalently, Algorithm A needs $K \approx 90$ to reach 95% detection while Algorithm B needs a single evaluation—nearly two orders of magnitude fewer model evaluations on this benchmark (Figure 3(a), Table 3). This gain is contingent on the harmful witness lying in the ShapG neighborhood (Theorem 3) and shrinks under correlation-matrix noise. Under Gaussian noise $\mathcal{N}(0, \sigma^2)$ on the correlation matrix (Figure 3(b)), Algorithm B degrades gracefully: detection at $K_{\text{adv}} = 5$ remains $1.000, 0.878, 0.652, 0.508$ for $\sigma = 0.05, 0.15, 0.30, 0.50$, each dominating Algorithm A's baseline (≤ 0.172).

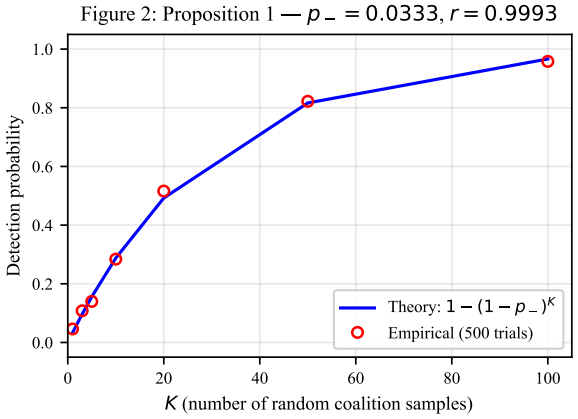


Figure 2: Empirical verification of Proposition 1 on an engineered voting game with $p_-^i = 0.0333$ (500 trials per K). Theory (solid line) and empirical rates (open circles) agree closely ($r = 0.9993$). For sparse harmful contexts ($p_- \ll 1$), Algorithm A requires $K \approx 90$ to reach 95% detection.

Figure 3: Algorithm B vs Algorithm A — §4.3.2

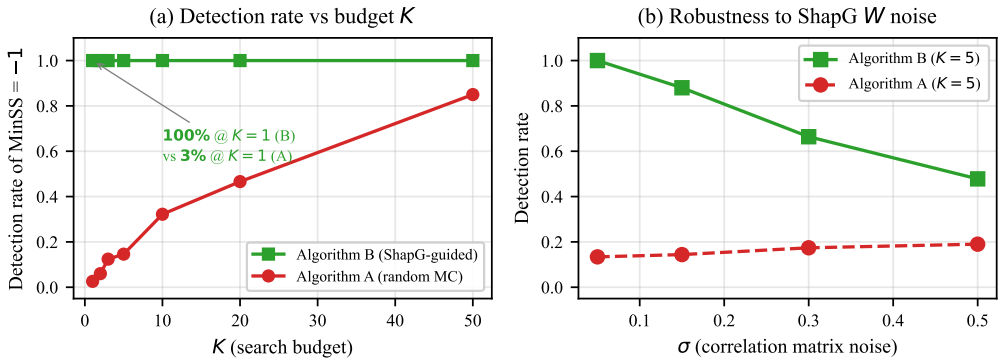


Figure 3: Algorithm B vs. Algorithm A on the engineered voting game of Section 4.2 ($p_-^i = 0.033$, 500 trials). (a) Detection rate vs. budget K: Algorithm B deterministically detects the harmful feature at $K = 1$; Algorithm A requires $K \approx 90$. (b) Robustness to Gaussian noise on the ShapG correlation matrix: even at $\sigma = 0.50$, Algorithm B detects the harmful feature in $> 50\%$ of trials, well above uniform sampling.

4.3.3 Hybrid Estimation Scheme

Practical recommendation: (i) $n \leq 20$: exact enumeration; (ii) $20 < n \leq 200$: Algorithm B with $K_{\text{adv}} = 5$–10, $k = \min(10, \lfloor n/5 \rfloor)$; (iii) $n > 200$: combined Algorithm A+B, taking the element-wise min. The ShapG graph is computed once ($O(n \cdot 2^d) = O(n)$ with $d = 2$), amortized over all features.

4.4 MOFS-MinPPI Algorithm

A feature subset is encoded as $\mathbf{m} \in \{0,1\}^n$. The three objectives are:

$$\min_{\mathbf{m} \in \{0,1\}^n} \left(\|\mathbf{m}\|_0,\ 1 - \text{AUC}(\mathcal{M}_\mathbf{m}),\ T_{\text{train}}(\mathcal{M}_\mathbf{m}) \right). \tag{8}$$

NSGA-II (Deb et al., 2002) performs the search. We replace uniform genetic operators with MinPPI-guided variants.

Table 3: Quantitative comparison of Algorithm A and Algorithm B (500 trials per setting).

(a) Detection of MinSS $= -1$

K	A detect	B detect	A $\bar{s}$	B $\bar{s}$
1	0.026	1.000	1.00	1.00
2	0.060	1.000	1.98	1.00
3	0.124	1.000	2.87	1.00
5	0.146	1.000	4.69	1.00
10	0.322	1.000	8.43	1.00
20	0.466	1.000	14.91	1.00
50	0.850	1.000	23.77	1.00

(b) Robustness to $\mathcal{N}(0, \sigma^2)$ **on** W **(**K=5**)**

σ	B detect	A detect
0.05	1.000	0.172
0.15	0.878	0.170
0.30	0.652	0.142
0.50	0.508	0.148

4.4.1 MinPPI-Guided Mutation

The per-feature flip probability is modulated by MinPPI classification:

$$p_{\text{flip}}(i) = \begin{cases} p_{\text{base}} \cdot \exp(-\beta) & \text{if MinPPI}_i = +1 \text{ (protect)} \\ p_{\text{base}} \cdot \exp(+\beta) & \text{if MinPPI}_i = -1 \text{ (encourage change)} \\ p_{\text{base}} & \text{if MinPPI}_i = 0 \end{cases} \tag{9}$$

with $\beta \geq 0$ controlling guidance strength. Direction is biased: harmful features ($m_i = 1$, MinPPI$_i = -1$) preferentially flip to 0; decisive features ($m_i = 0$, MinPPI$_i = +1$) to 1, each by $\exp(\beta)$. $\beta = 0$ recovers uniform mutation. The range $\beta \in \{0, 1, 2, 5\}$ is explored in Section 6.

4.4.2 MinPPI-Guided Crossover

Decisive features adjacent in G_{corr} (edge weight $> \tau = 0.3$) form *decisive groups*. Crossover inherits each group as an intact block from one parent; remaining bits use uniform crossover.

4.4.3 Complete Algorithm

MOFS-MinPPI operates in two phases. *Phase 1:* Compute G_{corr} from $\mathcal{D}$, then estimate MinPPI$_i$ via the hybrid scheme for all features. *Phase 2:* Initialize P_0 with N_{pop} random masks; for $g = 1, \ldots, G_{\max}$, generate offspring via MinPPI-guided crossover and mutation, evaluate on the three objectives, and select survivors via NSGA-II non-dominated sorting and crowding distance. Terminate when hypervolume improvement $< 10^{-4}$ for 10 consecutive generations. Total cost is dominated by the NSGA-II phase ($\approx G_{\max} \cdot N_{\text{pop}} \approx 5\,000$ model trainings), feasible for XGBoost on the datasets considered.

4.5 Discussion

4.5.1 Avoiding the WVG-FS Additivity Flaw

WVG-FS methods approximate $v(S) \approx \mathbb{1}[\sum_{i \in S} w_i \geq q]$, which is structurally incorrect for non-linear models (synergistic combinations undervalued, redundant ones overvalued). MOFS-MinPPI decouples three concerns: (i) v_α is evaluated by training $\mathcal{M}_S$—no additive surrogate; (ii) MinPPI serves only as soft guidance for variation operators; (iii) NSGA-II selection, based on true objectives, makes the final Pareto-optimality decision. Consequently, any MinPPI estimation error affects at most convergence speed, never Pareto-optimality—formalized as Proposition 2 below.

4.5.2 Spurious Interactions Under Feature Correlation

Rabitti & Borgonovo (2019) (Section 7.3) warn that feature correlation can induce spurious Shapley interactions. Our framework mitigates this at three levels: (i) binary thresholding via v_α discards fine-grained accuracy differences, absorbing minor estimation artifacts; (ii) the min operator evaluates every context, including those where correlated partners are absent, preventing false-positive

harmful classifications; (iii) G_{corr} makes correlation structure explicit and auditable. We do not claim to resolve the general spurious-interaction problem (Frye et al., 2020; Aas et al., 2021; Sundararajan & Najmi, 2020); rather, our design reduces its practical impact by asking a coarser, more robust question: "does adding i change whether the model meets the quota?" rather than "what is the exact interaction value?"

5 Theoretical Analysis

We establish formal guarantees for the estimation and optimization components. Proofs are deferred to the Appendix.

5.1 Sample Complexity of Algorithm A

Theorem 2 (Sample Complexity). Let $\mathcal{H} = \{i : \text{MinPPI}_i = -1\}$ and $p_{\min} = \min_{i \in \mathcal{H}} p_-^i > 0$. For confidence $\delta \in (0, 1)$, Algorithm A with

$$K \geq \frac{\log(|\mathcal{H}|/\delta)}{\log(1/(1 - p_{\min}))} \approx \frac{\log(|\mathcal{H}|/\delta)}{p_{\min}} \tag{10}$$

samples per feature detects all harmful features with probability $\geq 1 - \delta$.

Proof. For each $i \in \mathcal{H}$, Proposition 1 gives $\Pr[\text{miss } i] = (1 - p_-^i)^K \leq (1 - p_{\min})^K$. By the union bound, $\Pr[\text{any miss}] \leq |\mathcal{H}| \cdot (1 - p_{\min})^K$. Setting this $\leq \delta$ and solving for K yields equation 10; the approximation uses $-\log(1 - p_{\min}) \approx p_{\min}$ for small $p_{\min}$. □

Thus $K = \Omega(1/p_{\min})$: with $n = 500$ and $p_{\min} \approx 0.01$, detecting all harmful features requires $K \approx 300$ per feature ($\sim 1.5 \times 10^5$ trainings total). Empirical verification confirms theory matches observation ($r > 0.999$; Figure 2).

5.2 Sample Efficiency of Algorithm B

Theorem 3 (ShapG-Guided Efficiency). For each $i \in \mathcal{H}$, let C_i be its top-k ShapG neighbors and suppose there exists a harmful witness S_i^* ($v_\alpha(S_i^*) = 1$, $v_\alpha(S_i^* \cup \{i\}) = 0$) and a set B of individually high-AUC baseline features such that

$$S_i^* \subseteq C_i \cup B \;=:\; S_{\text{seed}}. \tag{11}$$

Suppose each adversarial perturbation independently retains every $j \in S_{\text{seed}}$ with probability $q_j \in (0, 1]$ (biased high for correlated neighbors) and excludes every $j \notin S_{\text{seed}}$. Let

$$\rho_i \;=\; \Big(\prod_{j \in S_i^*} q_j\Big)\Big(\prod_{j \in S_{\text{seed}} \setminus S_i^*} (1 - q_j)\Big) \;>\; 0. \tag{12}$$

Then Algorithm B with seed S_{seed} detects i within K_{adv} perturbations with probability $\Pr[\text{detection}] \geq 1 - (1 - \rho_i)^{K_{\text{adv}}}$. In the noiseless case ($S_{\text{seed}} = S_i^*$), the unperturbed seed is itself the witness, so detection is **deterministic** at $K_{\text{adv}} = 1$.

Proof. By equation 11, $S_i^* \subseteq S_{\text{seed}}$. A single perturbation reproduces exactly S_i^* when every $j \in S_i^*$ is retained (probability $\prod_{j \in S_i^*} q_j$) and every $j \in S_{\text{seed}} \setminus S_i^*$ is dropped (probability $\prod_{j \in S_{\text{seed}} \setminus S_i^*} (1 - q_j)$); these events are independent, so $\Pr[S = S_i^*] = \rho_i$. When $S = S_i^*$ the marginal is $v_\alpha(S_i^* \cup \{i\}) - v_\alpha(S_i^*) = -1$ and i is detected. Over K_{adv} independent perturbations, the probability of at least one hit is $1 - (1 - \rho_i)^{K_{\text{adv}}}$. If $S_{\text{seed}} = S_i^*$, the seed evaluated before any perturbation is already the witness, giving deterministic detection at $K_{\text{adv}} = 1$. □

Remark 3 (monotone-harm variant). If the harmful interaction persists on supersets of the witness within the seed—i.e. $v_\alpha(S) = 1 \wedge v_\alpha(S \cup \{i\}) = 0$ for all S with $S_i^* \subseteq S \subseteq S_{\text{seed}}$—then retaining S_i^* alone suffices irrespective of the other seed members, and the bound sharpens to $\Pr[\text{detection}] \geq 1 - (1 - \prod_{j \in S_i^*} q_j)^{K_{\text{adv}}}$. We do *not* assume this in general, since v_α is non-monotone (Remark 2); equation equation 12 keeps the exclusion factor for that reason.

The structural condition is empirically plausible: harmful features are typically redundant with correlated neighbors (C_i) in the presence of strong baseline features (B). Robustness experiments (Section 4.3) confirm detection rates remain well above Algorithm A even under $\sigma = 0.50$ noise.

5.3 Pareto Preservation Under Imperfect MinPPI

Proposition 2a (Selection Soundness, finite $G_{\max}$). Let $\widehat{\text{MinPPI}}$ be any (even adversarially incorrect) estimate. In each generation, MOFS-MinPPI selects survivors from the merged pool $\mathcal{R}_g = P_{g-1} \cup P_{\text{offspring}}$ by non-dominated sorting and crowding distance applied to the true objective $f(\mathbf{m}) = (\|\mathbf{m}\|_0,\ 1 - \text{AUC}(\mathcal{M}_\mathbf{m}),\ T_{\text{train}}(\mathcal{M}_\mathbf{m}))$ alone. Then (i) the selection outcome is independent of $\widehat{\text{MinPPI}}$; (ii) whenever the population size $N \geq |\mathcal{F}_1(\mathcal{R}_g)|$, the entire first non-dominated front of $\mathcal{R}_g$ survives, so no f-dominated solution displaces an f-non-dominated one; and (iii) the returned set P_{out} is mutually non-dominated under f.

Proof. $\widehat{\text{MinPPI}}$ enters only the generation operators (mutation equation 9, crossover), which determine *which* candidates populate $P_{\text{offspring}}$; once generated, a candidate is scored solely by f. Non-dominated sorting and crowding distance are functions of the f-values of $\mathcal{R}_g$ only, giving (i). The first front $\mathcal{F}_1(\mathcal{R}_g)$ ranks ahead of all dominated solutions; if $N \geq |\mathcal{F}_1(\mathcal{R}_g)|$ it is admitted in full, so dominated solutions can occupy only the remaining slots, giving (ii). At termination $P_{\text{out}} \subseteq \mathcal{F}_1(\mathcal{R}_{G_{\max}})$, whose members are pairwise non-dominated, giving (iii). □

Proposition 2b (Asymptotic Optimality). If $\beta < \infty$—so bit-flip mutation equation 9 assigns every $\mathbf{m} \in \{0,1\}^n$ positive probability each generation—and selection is elitist, then as $G_{\max} \to \infty$ the returned front converges almost surely to the true Pareto front P^*_{true}.

Proof. With $\beta < \infty$ the per-bit flip probability lies in $(0,1)$, so from any state the offspring distribution places positive probability on every point of $\{0,1\}^n$; the induced Markov chain over populations is therefore irreducible. Elitism guarantees that an f-non-dominated solution, once present, is never replaced by a solution it dominates (Prop. 2a(ii)). Irreducibility ensures every $\mathbf{m} \in P^*_{\text{true}}$ is generated in finite expected time and, being non-dominated, retained thereafter, while any dominated incumbent is displaced once a dominating point is sampled. Hence the front converges almost surely to P^*_{true}. The estimate $\widehat{\text{MinPPI}}$ alters the transition kernel but preserves its positivity, so it affects only the expected hitting times—the convergence speed—not the limit. □

Scope and implication. For finite $G_{\max}$ we guarantee only that P_{out} is internally non-dominated (Prop. 2a), *not* $P_{\text{out}} \subseteq P^*_{\text{true}}$; global optimality is the asymptotic statement (Prop. 2b). Within this scope MinPPI is a *safe* guidance signal: it shapes which candidates are proposed and how fast the front converges, while selection always defers to the true objective. Any importance method may thus be substituted without compromising the internal consistency of the returned front.

5.4 Summary

Table 4 consolidates the theoretical guarantees.

Author Contributions

If you'd like to, you may include a section for author contributions as is done in many journals. This is optional and at the discretion of the authors.

Acknowledgments

Use unnumbered third level headings for the acknowledgments. All acknowledgments, including those to funding agencies, go at the end of the paper.

Table 4: Summary of theoretical guarantees for the MinPPI-MOFS framework.

Result	**Practical consequence**
Lemma 1	MinPPI $\in \{-1, 0, +1\}$; no threshold tuning required.
Theorem 1	Clean semantic interpretation: decisive / harmful / neutral.
Lemma 2	MinSS = MinBz; use the computationally cheaper MinBz.
Prop. 1 / Thm. 2	Alg. A requires $K = \Omega(1/p_{\min})$; prohibitive for sparse harmful features.
Thm. 3	Alg. B near-deterministic when the witness lies in the ShapG neighborhood ($\sim$two orders of magnitude fewer evaluations).
Prop. 2a/b	MinPPI errors affect only convergence speed: finite-$G_{\max}$ front stays internally non-dominated, optimal as $G_{\max} \to \infty$.

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A Appendix

You may include other additional sections here.